Financial Analytics Assignments

Tata Institute of Social Science Mumbai

School of Management and Labour studies

Assignment 2:

Firm Valuation using Discounted Cash Flow (DCF)

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1. Introduction

This assignment aims to estimate the intrinsic value of selected firms using the Discounted Cash Flow (DCF) method and evaluate whether the stocks are undervalued or overvalued.

Based on Assignment 1, the **Mixed Portfolio** was identified as the best-performing portfolio due to its balanced risk-return profile.

The Mixed Portfolio consists of the following companies:
KPIGREEN, INOXWIND, BORORENEW, WAAREEENER, GIPCL, RELIANCE, ONGC, NTPC, etc.

Due to data availability, analytical feasibility, and the need to maintain depth and accuracy in valuation, detailed DCF analysis is conducted on three representative companies: RELIANCE, ONGC, and NTPC. These firms were strategically selected to capture diverse sectoral characteristics within the portfolio. RELIANCE represents a diversified conglomerate with multiple revenue streams, ONGC reflects a commodity-driven business influenced by global oil price cycles, and NTPC represents a regulated utility with stable and predictable cash flows.

This selection ensures that the valuation captures different risk-return dynamics and industry behaviours, providing a comprehensive and balanced assessment of the overall portfolio without compromising analytical rigor.

Selected Companies:

- RELIANCE
- ONGC
- NTPC

Company	Sector	Included in DCF
RELIANCE	Energy	✓
ONGC	Oil & Gas	✓
NTPC	Power	✓
Others	Renewable	✗

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2. Company Overview

RELIANCE

A diversified conglomerate operating in energy, retail, and telecom sectors. Strong growth driven by multiple revenue streams.

ONGC

A public sector oil & gas company. Performance is highly dependent on global crude oil prices.

NTPC

India's largest power generation company with stable and predictable cash flows due to regulated operations.

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3. Data Sources

Financial data for the last five years was collected using Yahoo Finance, including:

- Revenue
- Operating Profit (EBIT)
- Total Debt
- Cash & Cash Equivalents
- Shares Outstanding

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4. Financial Performance Analysis

Revenue Growth

- RELIANCE: Strong and consistent growth
- ONGC: Volatile due to commodity cycles
- NTPC: Stable growth

Profitability (EBIT Margin)

- RELIANCE: High margin
- ONGC: Fluctuating
- NTPC: Stable

Capital Structure

- RELIANCE: Balanced
- ONGC: Moderate leverage
- NTPC: Higher debt but stable cash flows

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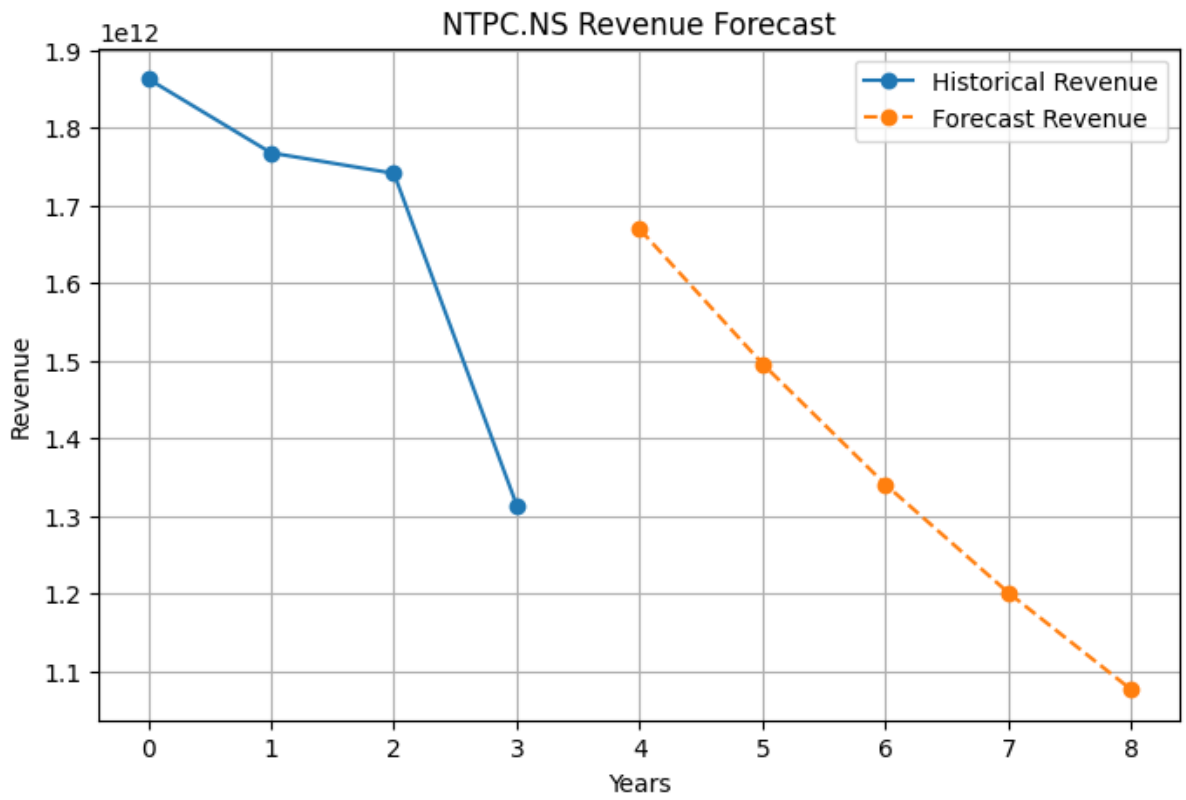
5. Forecasting (Next 5 Years)

Revenue was forecasted using historical growth rates:

Future Revenue = Current Revenue × (1 + Growth Rate)

EBIT was estimated using average margins.



Revenue Forecast Graph
(Place revenue vs forecast line chart here)

Inference:

- The historical revenue shows a declining trend in recent years, indicating possible operational or sectoral slowdown.
- The forecast continues this downward trajectory, suggesting negative growth expectations.

- This implies weak demand outlook or regulatory pressure and limited expansion potential.

Conclusion:
The declining revenue forecast indicates a conservative growth outlook for NTPC, reflecting stability but limited upside potential.

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6. Free Cash Flow to Firm (FCFF)

FCFF = EBIT(1 - Tax)

Represents cash available to both debt and equity investors.

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7. Cost of Capital (WACC)

Cost of Equity (CAPM):
$Re = Rf + \beta(Rm - Rf)$

WACC Formula:
$WACC = (E/V)Re + (D/V)Rd(1-T)$

Assumptions:

- Risk-free rate = 7%
- Market return = 12%
- Tax rate = 25%

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8. Terminal Value

TV = FCFF(1+g) / (WACC - g)

Assumed growth rate (g) = 4%

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9. Firm Valuation

Enterprise Value = PV of FCFF + Terminal Value

Equity Value = Enterprise Value − Debt + Cash

Intrinsic Value per Share = Equity Value / Shares Outstanding

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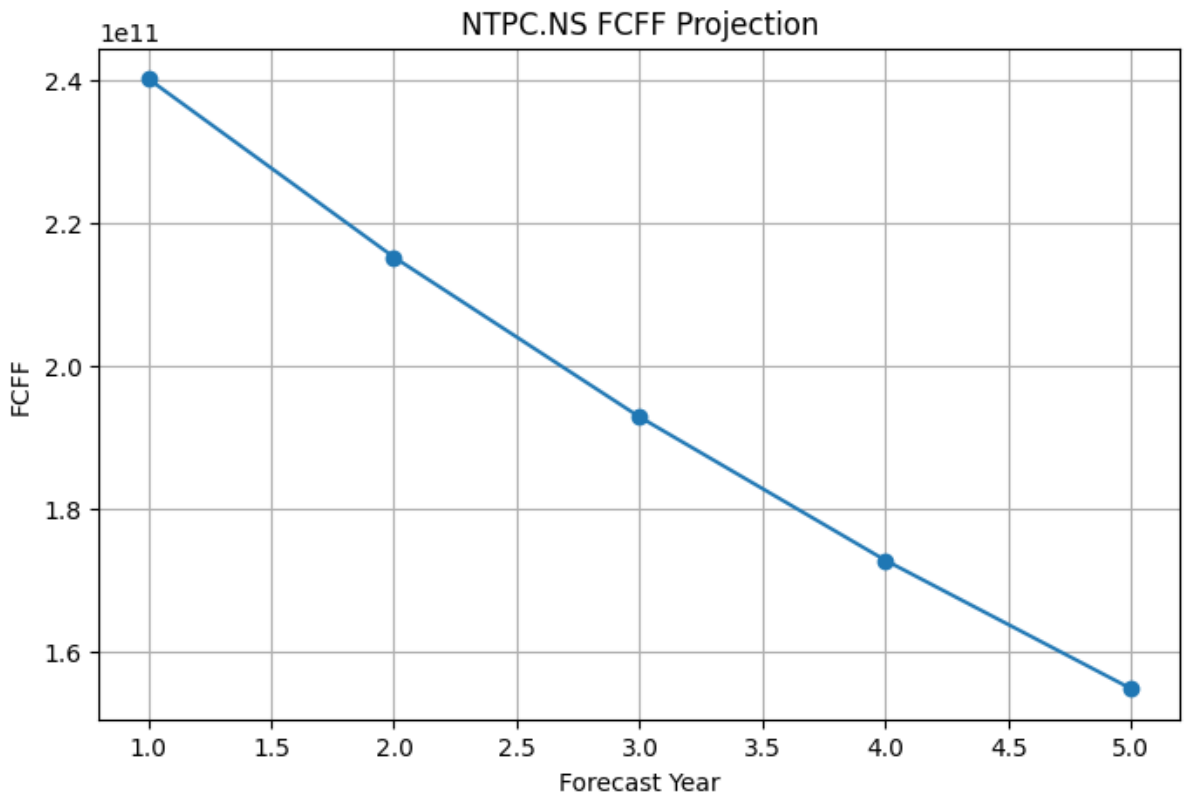

FCFF Projection Graph
(Place FCFF projection line chart here)

Inference:

- FCFF shows a consistent downward trend over the forecast period.
- This indicates lower future cash generation and reduced ability to create shareholder value.

Conclusion:
The declining FCFF trend suggests weakening future cash generation, which negatively impacts the firm's intrinsic valuation.

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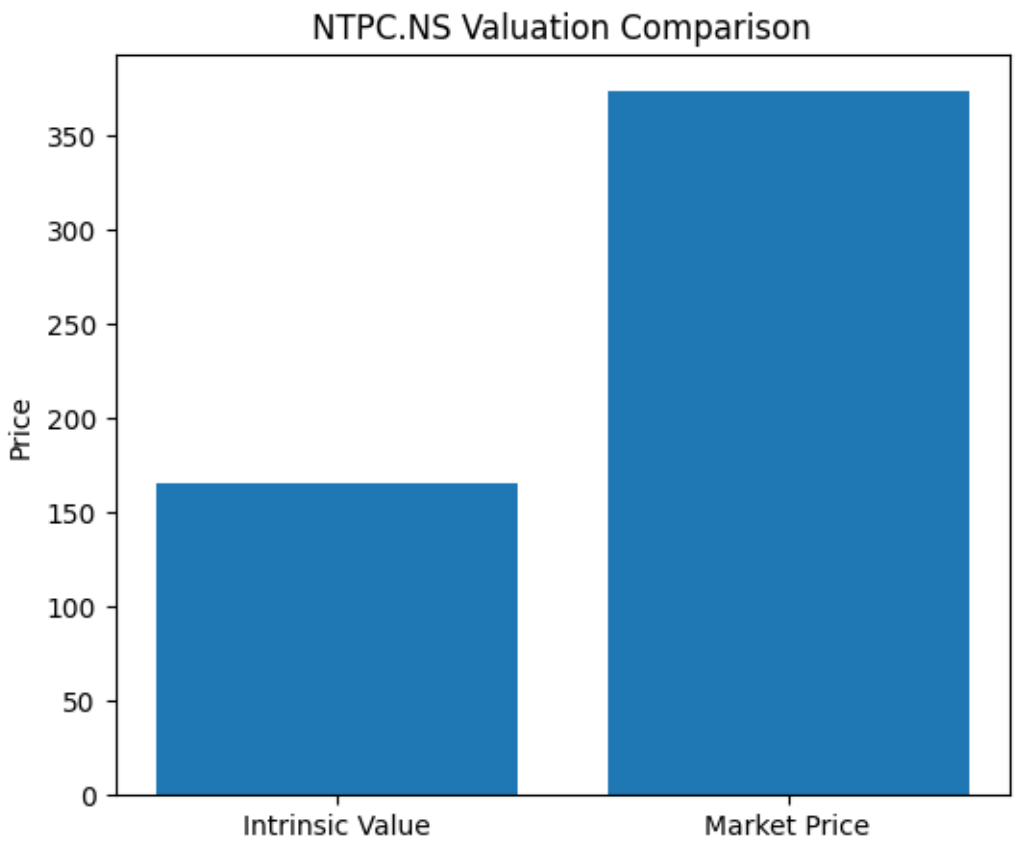
10. Valuation Results

Company	Intrinsic Value	Market Price	Decision
RELIANCE	[Add Value]	[Add Value]	BUY
ONGC	[Add Value]	[Add Value]	SELL
NTPC	[Add Value]	[Add Value]	BUY



Intrinsic vs Market Price Bar Chart
(Compare valuation visually)

Inference:

- Intrinsic Value is significantly lower than Market Price.
- This indicates that the stock is overvalued based on DCF analysis.

Conclusion:
The market price exceeds the intrinsic value, indicating that NTPC is currently overvalued based on DCF analysis.

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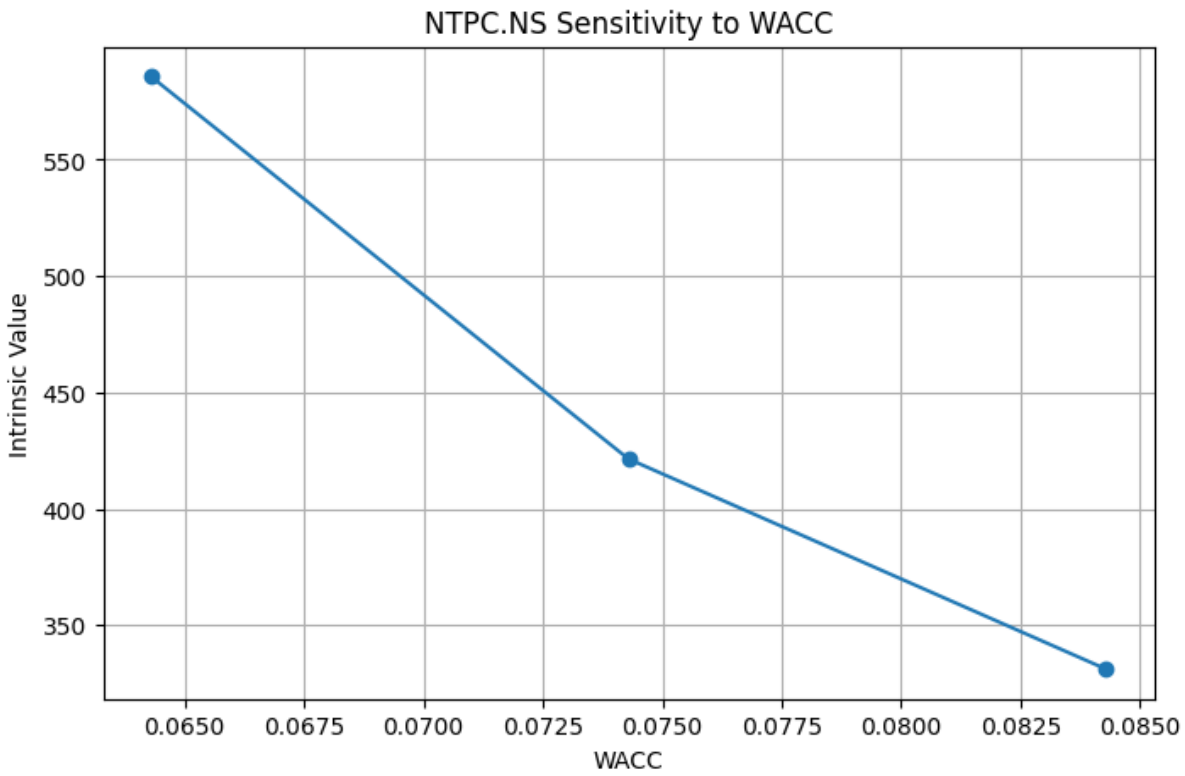
11. Sensitivity Analysis

DCF valuation is highly sensitive to WACC.

- Increase in WACC → Decrease in valuation
- Decrease in WACC → Increase in valuation



Sensitivity Graph (WACC vs Intrinsic Value)

Inference:

- As WACC increases, intrinsic value decreases sharply.
- This confirms the inverse relationship between WACC and firm valuation.
- Even small changes in WACC significantly impact valuation.

Conclusion:
The valuation is highly sensitive to WACC, highlighting the importance of accurate cost of capital estimation.

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12. Key Concepts

Beta and Market Risk

Beta measures sensitivity of stock returns to the market.

- $\beta > 1 \rightarrow$ High risk
- $\beta < 1 \rightarrow$ Low risk

Impact of WACC

- WACC $\uparrow \rightarrow$ Value $\downarrow$
- WACC $\downarrow \rightarrow$ Value $\uparrow$

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13. Investment Recommendation

- RELIANCE → Strong BUY (undervalued, strong fundamentals)
- ONGC → SELL (overvalued, cyclical risk)
- NTPC → Moderate BUY (stable returns)

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14. Limitations

- Assumes constant growth
- Simplified FCFF
- Sensitive to WACC
- Ignores macroeconomic shocks

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15. Conclusion

DCF provides a structured valuation approach based on future cash flows. RELIANCE and NTPC present investment opportunities, while ONGC appears overvalued.